

P–T Pseudosections via pypsbuilder

Module 2 — Practical Thermodynamic Modelling for Petrology

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Objective

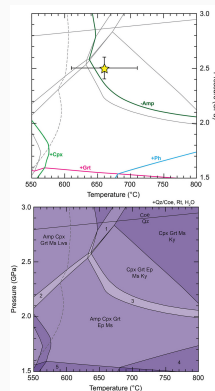
Automate P–T phase diagram (pseudosection) construction, univariant-line mapping, and post-processing with `pypsbuilder`.

- Automated THERMOCALC setup
- `ptbuilder` execution
- `psexplorer` visualization

Introduction

How Do We Study Metamorphic Rocks?

- **Conventional thermobarometry** (*inverse modelling*): use mineral compositions in a rock to retroactively estimate the P–T conditions of formation — a single point, with uncertainty
- **Phase equilibrium modeling** (*forward modelling*): use the rock's bulk composition to *predict* which minerals should be stable at any P–T condition — a full P–T map



Top: thermobarometric P–T estimate with error bars. Bottom: pseudosection for the same bulk composition — the full assemblage map.

Why Phase Equilibrium Modeling?

Rests on Powell's (1978) equilibrium premise: at least some features in metamorphic rocks record the achievement of chemical equilibrium, at some scale, at some time — which is why predicting the state of minimum Gibbs free energy is worth doing.

- **Predicts mineral stability & composition:** which minerals are stable, and what they're made of, at any P–T
- **Traces P–T paths:** compare observed assemblages and zoning to the modeled diagram to reconstruct a rock's burial–heating–exhumation history
- **Links to geodynamics:** connects mineral-scale observations to crustal-scale processes — subduction, collision, exhumation

Chemical Systems & Components

A **component** is one of the minimum set of chemically independent oxides needed to describe the composition of every phase in the system.

- **KFMASH** — K_2O - FeO - MgO - Al_2O_3 - SiO_2 - H_2O : the classic simplified pelitic system
- **NCKFMASHTO** — adds Na_2O , CaO (plagioclase, white micas), TiO_2 , ferric O (ilmenite, rutile, magnetite)
- **MnNCKFMASHTO** — adds MnO , critical for garnet nucleation and subsolidus stability

This is the system you're already using

The oxide set in this deck's THERMOCALC bulk composition (H_2O SiO_2 Al_2O_3 CaO MgO FeO K_2O Na_2O TiO_2 MnO O) is MnNCKFMASHTO.

The Gibbs Phase Rule

For a system at chemical equilibrium, the degrees of freedom (**variance**, F) obey:

$$F = C - P + 2$$

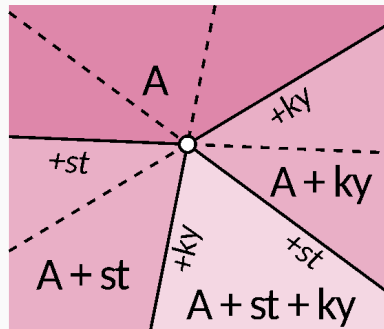
C = number of components, P = number of coexisting phases.

Topology	F	P	On a P–T diagram
Invariant point	0	$C + 2$	a point
Univariant line	1	$C + 1$	a line
Divariant field	2	C	a 2-D field
Multivariant	> 2	$< C$	a wide, higher-order field

This is the vocabulary invariant-point topology analysis and this afternoon's worked example both build on.

Topology at an Invariant Point

- **Coincidence Theorem**
(Morey–Schreinemakers): each reaction curve has a **stable** segment ending at the invariant point and a **metastable** continuation beyond it
- **Metastable Extension Rule**: that continuation must project into the sector (divariant field) where the reaction's missing phase is stable
- **Zen's 180° Rule**: no single phase or assemblage occupies more than half the arc around the point



Zero-mode univariant lines ($st=0$, $ky=0$) meeting at an invariant point, bounding the higher-variant fields

ZPF lines stop at the invariant point

In a **petrogenetic grid**, univariant curves are drawn continuously through the invariant point as dashed metastable extensions. In a **pseudosection**, univariant boundaries are **ZPF (zero-phase-fraction) lines** that instead **stop** at the invariant point — Gibbs energy minimization for the fixed bulk composition simply excludes the metastable continuation, even though the same coincidence/extension rules still determine where it would be allowed to end.

This afternoon's Core Workflow works through a full kyanite–staurolite example by hand; ptbuilder's **Search/Dogmin** automate exactly this.

Definitions & Fundamental Concepts

Phase Diagram (General)

A general thermodynamic map depicting the equilibrium stability of mineral assemblages across physical (P , T) or chemical variables.

Petrogenetic Grid (P - T Projection)

A phase diagram mapping **every reaction possible** in a chemical system, **independent of any one bulk composition** (W) — the full topology of invariant points and univariant reactions.

Pseudosection (Isochemical Phase Diagram)

A phase diagram calculated for a **single, fixed whole-rock bulk composition** (W) — filters the grid down to only what that composition can reach.

- **Phase Rule Basis:** Governed by Gibbs Phase Rule $F = C - P + 2$.
- **Forward Modelling:** Pseudosections isolate only the stable assemblages accessible to a particular rock specimen.

Why P–T Pseudosections?

Unlike a generic phase diagram, a pseudosection is calculated for one **specific rock's bulk composition**.

- Defines the stable mineral assemblage at every P–T point
- Comparing it to observed assemblages reveals the rock's P–T evolution
- Predicts not just *which* minerals, but their **modal abundances and compositions**

Bridge from Module 1

The bulk composition you exported with `petropandas` this morning is exactly what a pseudosection needs as input.

Pseudosections are one member of a wider diagram family: P–T projections, compatibility diagrams, contoured pseudosections (isopleths), and T–X/P–X sections — all covered in this module.

Key Comparison

Feature	General Phase Diagram	Pseudosection
Bulk Composition	System-wide (all potential rocks in model system, e.g. KFMASH)	Fixed (one specific whole-rock chemistry)
Primary Focus	Invariant points ($F = 0$) and univariant lines ($F = 1$)	Multivariant fields ($F \geq 2$) accessible to sample
Field Boundaries	Univariant reactions	“Zero-mode” lines (phase abundance drops to 0)
Practical Utility	Generic topological reference frame	Matches petrographic modes/compositions to P - T

System-Wide vs. Sample-Specific Modelling

Petrogenetic Grids (P - T Projections)

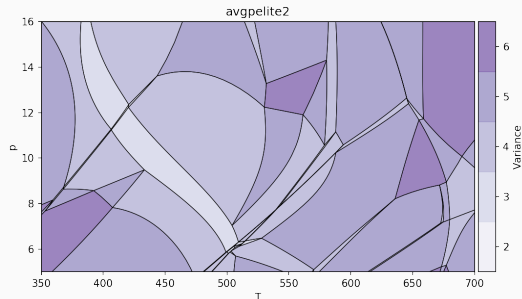
- Displays **all possible reactions** in a chemical system.
- Bulk-independent reference topology.
- Can become extremely complex and cluttered in large multi-component systems ($C \geq 10$).

Pseudosections

- **Filters out** reactions inaccessible to the rock composition.
- Directly links observed mineral modes and microprobe data to P - T conditions.
- Essential for reconstructing P - T - t - X evolution paths.

Reading a Pseudosection

- **Shading & Variance**
($F = C - P + 2$):
 - **Lighter fields** = lower variance
(more coexisting minerals)
 - **Darker fields** = higher variance
(fewer coexisting minerals)
- **Nodes & Lines:** boundaries are zero-mode lines meeting at nodes where two phase modes go to zero simultaneously



Fields shaded by variance — lighter = lower variance

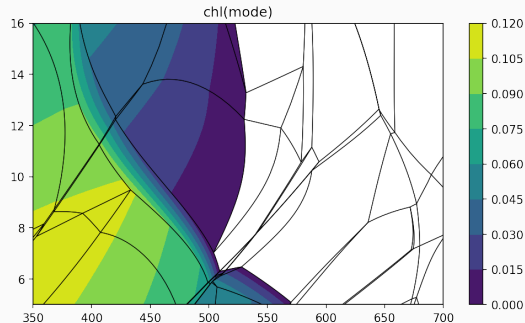
Isopleth Contouring

Contouring the Diagram

Pseudosections can be contoured with lines of constant:

- Mineral modal abundances (vol% or mol%)
- Mineral composition parameters (e.g., X_{Mg} , X_{Ca} , Si cations p.f.u.)

Allows precise quantitative thermobarometry.



Chlorite modal-abundance isopleths over the topology

Available Tools for Pseudosection Modeling

All four solve the same problem — stable assemblage vs. P–T for a bulk composition, on the same Holland & Powell family of internally-consistent databases — but trade off speed against boundary precision very differently.

Tool	Core Method	Speed	Boundary Precision	Primary Strength
THERMOCALC	Direct boundary calc.	Slow / manual	Exact, smooth curves	Topological precision & pedagogy
Perple_X	Global Gibbs min. (grid)	Very fast	Approximated, grid artifacts	Rapid exploratory modeling
Theriak-Domino	Adaptive Gibbs minimization	Moderate	High, clean boundaries	Robust automated minimization
MAGEMin	Parallel C/Julia solver	Ultra fast (parallel)	High	Geodynamic integration & scaling (<i>Module 3</i>)

Tool Trade-offs: Pros & Cons

THERMOCALC

- + Exact, smooth boundaries; enforces strict thermodynamic rules
- Labor-intensive; manual field-by-field setup, needs known stable phases

Theriak-Domino

- + Adaptive grid gives clean boundaries without manual setup; very stable algorithm
- Slower than Perple_X on dense grids; dated GUI; solution-model edits need code changes

Perple_X

- + Full pseudosection in minutes; wide range of databases and custom activity models
- Pseudocompound discretization causes pixelation artifacts and can miss narrow phase fields

MAGEMin

- + Multithreaded — unmatched speed; native Julia / MATLAB / geodynamic-code interfaces
- Newer, smaller ecosystem; fewer pre-packaged solution models than Perple_X

Our Recommendation: THERMOCALC, Minus the Pain

For publication-grade diagrams

When a paper or thesis figure needs **exact, topologically correct** P–T boundaries, THERMOCALC remains the reference standard — no pseudocompound grid artifacts to explain away.

`pypsbuilder` eliminates most of THERMOCALC's classic drawbacks:

- **Search & Dogmin** automate invariant-point location — no more guessing which phases are stable first
- GUI-driven **Calc P / Calc T** replace hand-edited scriptfiles for every single boundary
- `psexplorer` adds post-processing THERMOCALC lacks: gridded isopleths, phase modes, and P–T paths

Bottom line: exact THERMOCALC topology, at a workflow speed much closer to Perple_X or MAGEMin.

What Should You Choose?

“Ease may be quick, but depth comes with effort.”

This workshop gives you both ends of that trade-off:

- **MAGEMin** (Module 3) — fast, automatic, great for exploring
- **THERMOCALC via pypsbuilder** (this module) — slower, hands-on, forces you to understand *why* the diagram looks the way it does



Core Takeaway

A general phase diagram provides the **theoretical map of the system**, while a pseudosection provides the **specific trajectory of a given rock**.

Golden Rules for Application

1. **Petrography First:** Pseudosections are meaningless without careful thin-section observations and equilibrium texturing.
2. **Bulk Precision:** Account for problematic components (e.g., apatite correction for Ca, H₂O saturation, Fe³⁺ ratio).
3. **Equilibrium Volume:** Ensure the bulk chemistry matches the actual volume of equilibration.

What is pypsbuilder?

What is pypsbuilder?

Elevator pitch

A **PyQt front-end for THERMOCALC** that automates construction and exploration of P–T (and T–X, P–X) pseudosections: it drives THERMOCALC to calculate univariant (ZPF) lines and invariant points, tracks their topology, and provides post-processing tools (isopleths, phase modes, P–T paths) for the finished diagram.

No more hand-drawing curve by curve — but you still need to understand the Gibbs Phase Rule and invariant-point topology.

Installation

Requires Python $\geq$ 3.10. Installed this morning as part of one shared uv project together with petropandas (Module 1):

```
uv add petropandas[lab] pypsbuilder[pyqt6]
```

pypsbuilder is a *front-end*: THERMOCALC itself must also be set up (next slide). Full details, including the from-scratch/downloaded-materials install options, are in the workshop's `install.md`.

THERMOCALC Setup: `tcinit`

Automated (recommended for the workshop)

In an empty working directory:

```
uv run tcinit
```

- Downloads THERMOCALC 3.51 for the current platform
- Lets you pick a bundled dataset / a-x file: `metapelite`, `metabasite`, `igg25`, `igg24`
- Writes scriptfiles already tagged for `ptbuilder`/`txbuilder`/`pxbuilder`

Manual alternative: prepare a directory with the THERMOCALC/`drawpd` executables, preferences file, dataset, a-x file, and insert the `%{PSBGUESS...}` / `%{PSBCALC...}` / `%{PSBBULK...}` tags by hand (see `examples/avgpelite` in the repository).

THERMOCALC Bulk Composition

Use petropandas to format an EPMA/whole-rock bulk composition for THERMOCALC directly from the workshop's own data:

```
from petropandas import pd

df = pd.read_excel("data/avg_pelite/microprobe_data.xlsx", sheet_name="Bulk", index_col=0)
df.bulk.TCbulk(H2O=4, oxygen=0.15)
```

Output:

```
bulk H2O SiO2 Al2O3 CaO MgO FeO K2O Na2O TiO2 MnO O
bulk 13.269 61.543 10.735 1.266 3.229 5.353 2.198 1.527 0.630 0.100 0.150 % avgpelite
```

Copy the output line, run `uv run ptbuilder`, open the **Script file** tab, and replace the bulk composition inside the `%{PSBBULK-BEGIN/END}` block — then **Save**.

Core Workflow

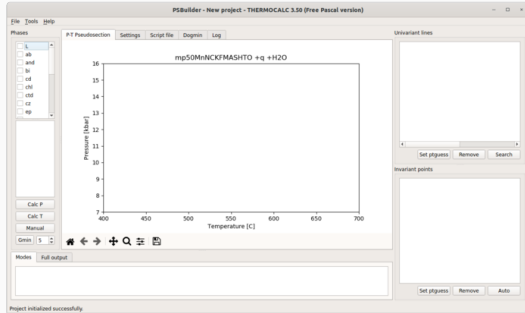
Tool	Role
tcinit	set up THERMOCALC + dataset
ptbuilder / txbuilder / pxbuilder	interactive GUI builders
psexplorer (PTPS/TXPS/PXPS)	Python post-processing API
psshow / psgrid / psiso	command-line visualization

Typical path: tcinit → ptbuilder (build topology) → psexplorer (grid & visualize).

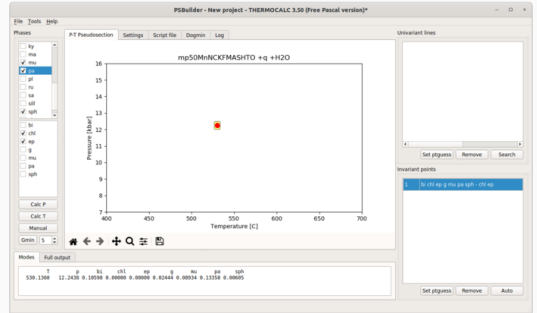
1. Building the Diagram — ptbuilder

- Create a new project: *File* → *New*, point at the THERMOCALC working directory (auto-validates settings and phases)
- **Calc P / Calc T** — calculate univariant lines
- **Search** — locate invariant points
- **Dogmin** — find stable assemblage at a point when the topology is ambiguous
- Manual point/line entry when needed
- **Build areas** (Ctrl-A) to verify the finished topology

pypsbuilder in Action: the Builder GUI

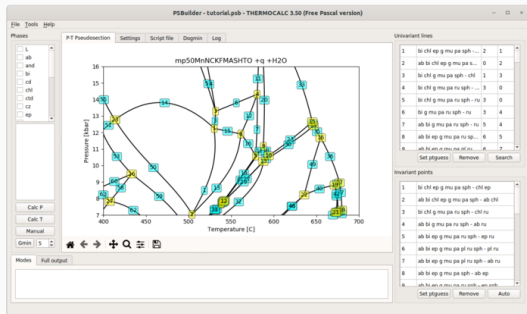


New project / initial setup

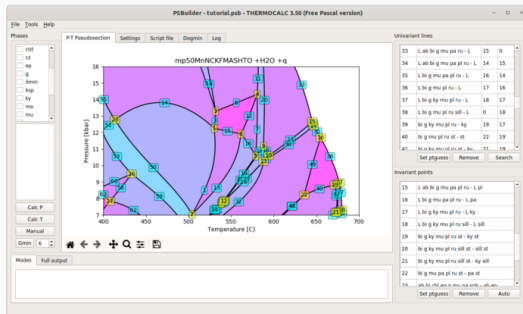


Invariant point search

From Topology to Finished Diagram



Completed univariant-line topology



Areas built (Ctrl-A) — topology check

2. Launching the Builder

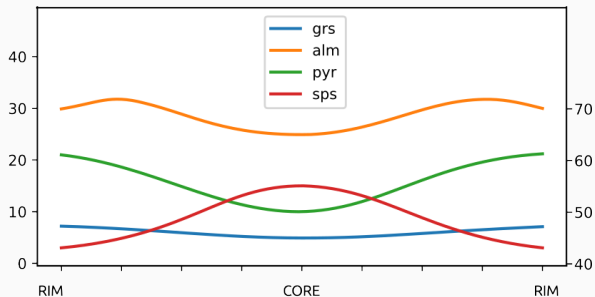
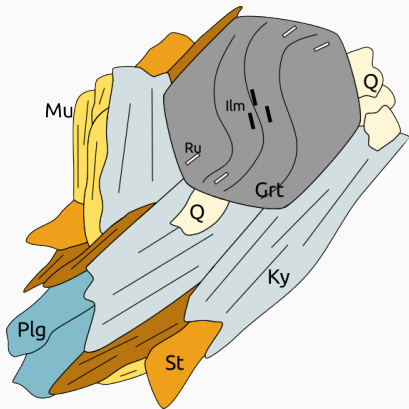
```
uv run ptbuilder
```

Same idea for T–X and P–X sections:

```
uv run txbuilder  
uv run pxbuilder
```

How to start?

The most critical part is to start your calculations with a correctly selected stable assemblage. There are plenty of ways to get it. Start with petrography, check already published pseudosections with similar composition, use `Perple_X` or `MAGEMin`, use `dogmin`, etc.



Si in Mu 3.05-3.06
xMg in St 0.26-0.32
xMg in Bi 0.22-0.26

Invariant Point Topology - kyanite-staurolite assemblage

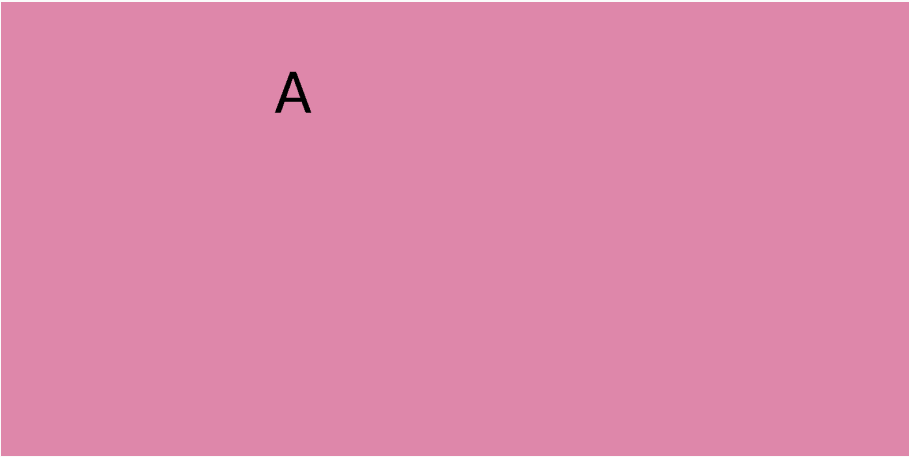
In P - T pseudosections (fixed bulk composition diagrams) for complex systems like MnNCKFMASHTO, the same coincidence and metastable-extension rules govern how zero-mode **univariant (ZPF)** lines ($M_i = 0$, representing the appearance/disappearance boundaries of phases) intersect at **invariant** points.

When staurolite (st) and kyanite (ky) both reach zero modal abundance ($M_{\text{st}} = 0$ and $M_{\text{ky}} = 0$) at a single point, they form a pseudosection **invariant** point within a coexisting stable assemblage A (bi + g + mu + pl + q + ru + H₂O).

Field	Phase Assemblage	Modal Status
Field 1	A	$M_{\text{st}} = 0, M_{\text{ky}} = 0$
Field 2	A + St	$M_{\text{st}} > 0, M_{\text{ky}} = 0$
Field 3	A + Ky	$M_{\text{st}} = 0, M_{\text{ky}} > 0$
Field 4	A + St + Ky	$M_{\text{st}} > 0, M_{\text{ky}} > 0$

Phase stability fields

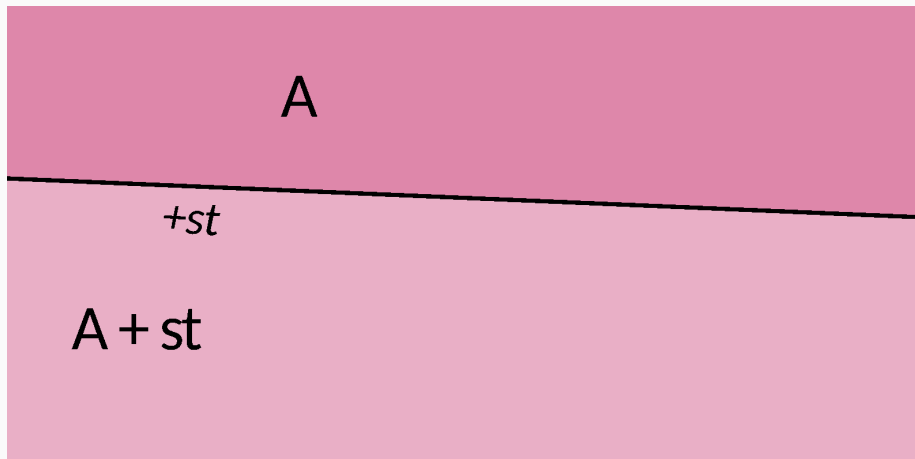
Here we start with phase stability field A with assemblage **H₂O bi g mu pl4tr q ru**



A

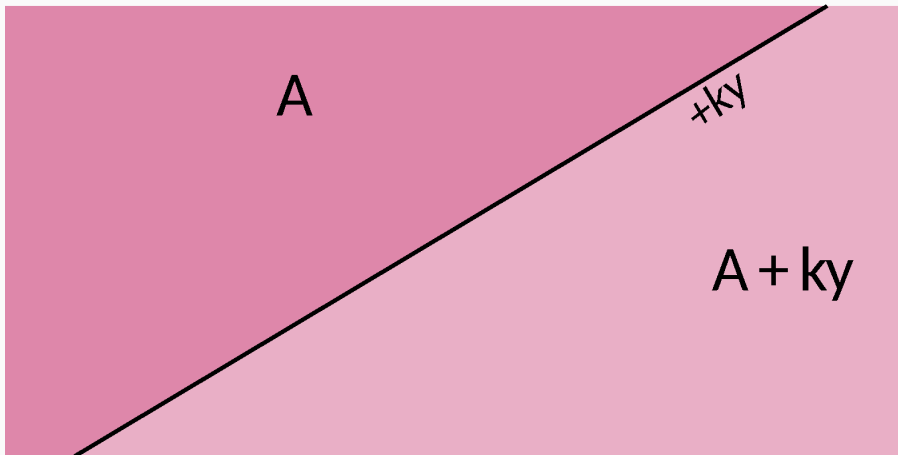
Univariant line 1

In lower pressure, appearance of staurolite defines a new lower variance field



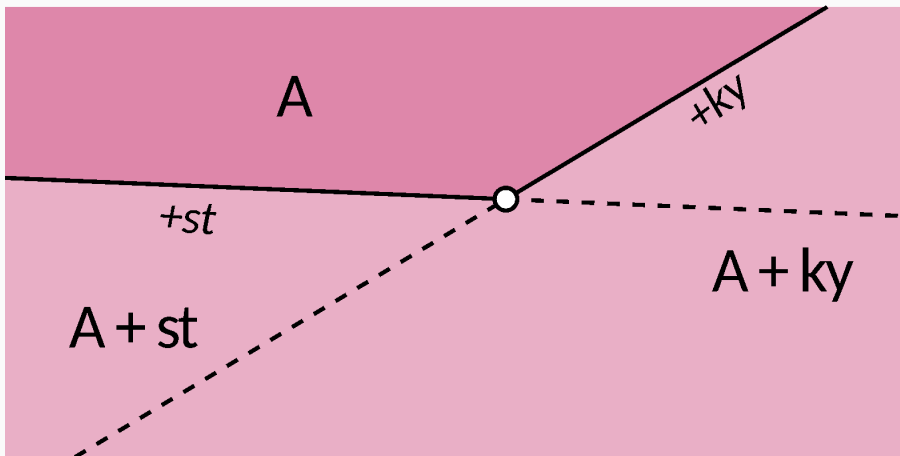
Univariant line 2

In higher temperature appearance kyanite also define a new lower variance field



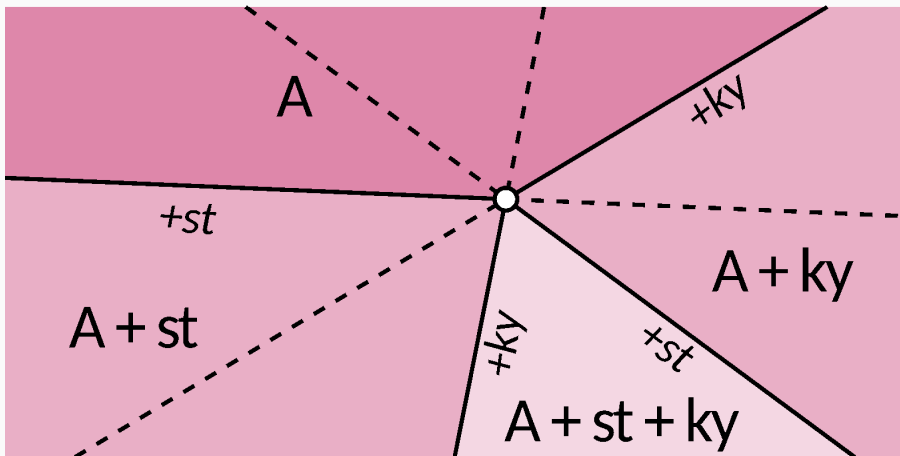
Stable and metastable parts

Two univariant lines define invariant point separating metastable part



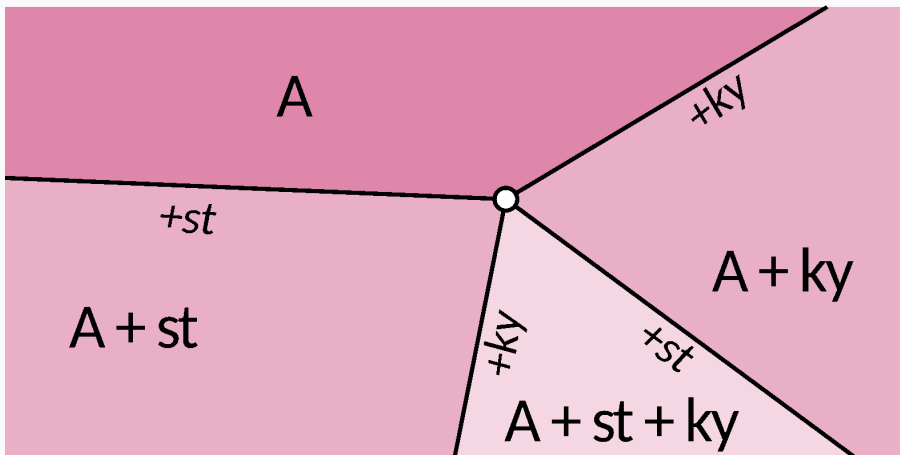
Remaining univariant lines to fulfill phase rule

Other two univariant lines resolve phase rule



Final topology of univariant reaction curves radiating from an invariant point

Governed by the Coincidence Theorem, the Extension Rule, and Zen's 180° Rule



3. Post-Processing — psexplorer

- Load a finished builder project (.ptb file)
- **Grid** compositions across the diagram
- Show the pseudosection with labelled fields
- Identify the stable assemblage at any P–T point
- Draw **isopleths** (composition contours)
- Calculate **P–T paths** through the diagram

Available as a Python API (PTPS/TXPS/PXPS) or as CLI scripts: psshow, psgrid, psiso.

```
# draw the finished pseudosection with areas
psshow -b /path/to/project.ptb

# calculate a compositional grid
psgrid /path/to/project.ptb --nx 50 --ny 50

# draw isopleth contours for garnet Xca
psiso /path/to/project.ptb -f g -e mode --step 0.005 \
  --high "H2O bi g mu pa pl q ru" --out chl ep --cmap YlGnBu_r
```

```
from pypsbuilder import PTPS

pt = PTPS('/path/to/project.ptb')

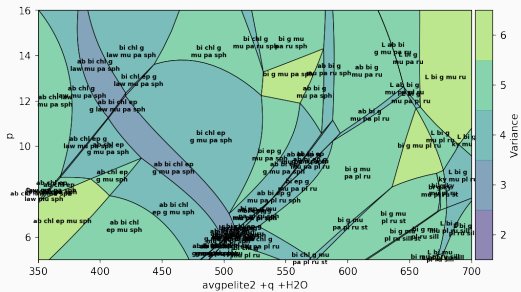
if not pt.gridded:
    pt.calculate_composition(nx=50, ny=50)

pt.show(cmap='viridis', bulk=True, label=True)

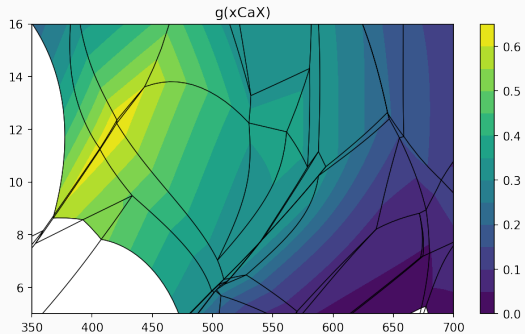
key = pt.identify(550, 13) # assemblage at 550C, 13 kbar
pt.isopleths('g', 'xCaX', N=14)
```

Explorer Output

H ₂ O	SiO ₂	Al ₂ O ₃	CaO	MgO	FeO	K ₂ O	Na ₂ O	TiO ₂	MnO	O
50	35.6	5.8	0.695	2.33	2.89	1.34	0.495	0.335	0.05	0.005

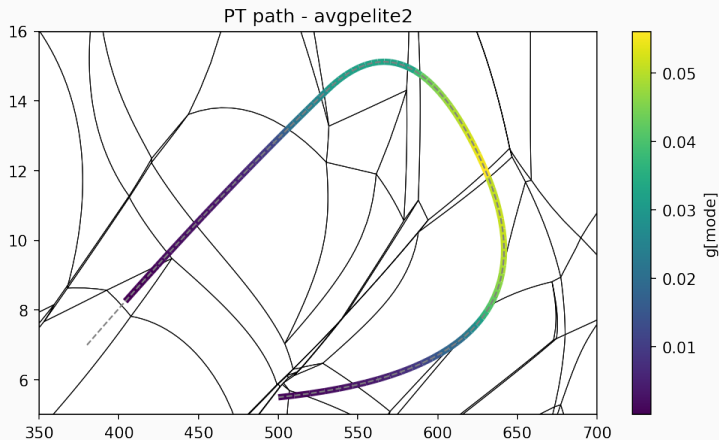


Labelled pseudosection (.show())



Isopleth contours (.isopleths())

P-T Path Calculation



From a compositional/textural sequence, psexplorer can reconstruct a P-T path through the gridded pseudosection.

Hands-on

1. Run `tcinit` and pick the `metapelite` dataset
2. Use the bulk composition exported from Module 1 (or the bundled `examples/avgpelite` project)
3. Launch `ptbuilder`, locate the first invariant point with **Search**, then calculate the surrounding univariant lines
4. Build areas (`Ctrl-A`) once the topology looks complete
5. Load the project into PTPS, grid it, and draw one isopleth

Recap

Key takeaways

- `tcinit` removes the pain of manual THERMOCALC setup
- `ptbuilder` automates invariant-point search and univariant-line calculation
- `psexplorer` turns a finished project into publication-quality figures — and P–T paths

Resources:

- Source: github.com/ondrolexa/pypsbuilder
- Example projects: `examples/avgpelite`, `examples/avgpelite_34`

Next: Fast Gibbs Minimization with MAGEMin